

SentinelNet – AI-Powered Network Intrusion Detection System (NIDS)

1. Introduction

SentinelNet aims to build a **smart Network Intrusion Detection System (NIDS)** powered by **AI and machine learning**.

The central goal is to automatically **identify and respond to suspicious or potentially harmful network traffic and cyber-attacks in real time**.

Using historical data, machine learning models will be trained to:

- Distinguish between normal and malicious activities
- Extract crucial features
- Classify threats
- Trigger alerts to help network defenders respond quickly and effectively

2. Literature

The field of **Intrusion Detection Systems (IDS)** has evolved from rule-based methods to AI-powered frameworks. Literature highlights:

- **Signature-Based IDS:** Relies on predefined attack signatures, effective for known threats but fails on zero-day attacks.
- **Anomaly-Based IDS:** Uses statistical models and ML to detect unusual traffic patterns, can detect unknown threats but may have higher false positives.

Benchmark datasets shaping IDS research:

- **KDD Cup 1999:** Early dataset, criticized for redundancy and outdated attacks.
- **NSL-KDD:** Improved version, duplicates removed, balanced dataset for academic evaluation.
- **CICIDS2017:** Realistic enterprise traffic, modern attacks like DDoS, infiltration, and web exploits.

ML algorithms widely used:

- Decision Trees, Random Forests, SVMs, Neural Networks
- Deep learning, ensemble methods, hybrid IDS models

Insights:

- Feature selection and preprocessing are critical.
- Class imbalance is a challenge.
- Hybrid approaches combine speed of signature-based and intelligence of anomaly-based IDS.

3. Methodology

3.1 Database

3.1.1 NSL-KDD Dataset:

A refined dataset to reduce redundancy:

- 41 features: basic (protocol, service, bytes) + traffic-level (flags, errors)
- Training: 125,973 records
- Test: 22,544 records
- Attack types:
 - DoS: smurf, neptune

- Probe: portsweep, nmap
- U2R: buffer_overflow, rootkit
- R2L: ftp_write, guess_passwd

3.1.2 CICIDS2017 Dataset:

Realistic enterprise network traffic:

- 78–83 features: flow, statistical, packet-level
- Over 2.8 million labeled flow records across 5 days
- Attack types: DDoS (Hulk, GoldenEye, Slowloris), Brute Force, Web Attacks (SQLi, XSS), Botnet/Infiltration, Reconnaissance (PortScan, Heartbleed)

3.1.3 Dataset Comparison:

Attribute	NSL-KDD	CICIDS2017
No. of Features	41	78–83
Total Records	~150,000	>2.8 million
Traffic	Simulated, old-generation	Realistic, enterprise-scale
Usage	Lightweight model evaluation	Practical deployment
Attack Variety	4 legacy categories	14+ modern attacks

3.2 Preprocessing

- Handling missing values
- Feature scaling: MinMaxScaler, StandardScaler
- Encoding categorical features: One-Hot Encoding
- Stratified train-test split to maintain class balance

3.2.1 Scaling vs Normalization

Aspect	Scaling (Standardization)	Normalization (Min-Max)
Definition	Transforms features to have mean=0 and standard deviation=1	Rescales features to a fixed range, usually [0, 1]
Purpose	Handles varying feature scales for better ML performance	Ensures features are comparable and bounded for algorithms like Neural Networks
Use Cases	Algorithms assuming Gaussian distribution (SVM, Logistic Regression)	Algorithms sensitive to feature bounds (NNs, image data)
Effect on Outliers	Sensitive to outliers (outliers get compressed between 0 and 1)	Less sensitive, but outliers can influence mean and std
Preserves Data Shape	Yes, but scales features to range	Yes, but centers around mean 0
When to Use	Useful for algorithms requiring bounded input (e.g., Neural Networks, KNN)	Useful for algorithms assuming Gaussian distribution (e.g., Logistic Regression, SVM, PCA)

3.3 Novelty

- Hybrid detection combining anomaly-based intelligence with signature-based methods
- Advanced preprocessing and feature selection for higher performance
- Class imbalance handling using SMOTE and weighted losses

3.4 Proposed Method

Pipeline:

- Data loading and cleaning
- Feature scaling and encoding
- Model training: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting

- Evaluation: Confusion Matrix, Accuracy, Precision, Recall, F1-score, ROC-AUC

3.4.1 Why AI is Critical in Cybersecurity?

Cybersecurity is one of the most urgent challenges today, with threats growing rapidly in both complexity and scale.

Traditional rule-based methods struggle against evolving attacks, but AI enables:

- Detecting subtle attack patterns in massive data streams
- Continuous learning from historical + new data
- Identifying zero-day threats never seen before
- Automating detection and response for proactive defense

3.4.2 What Excites Me About Building SentinelNet

- Working with advanced datasets and algorithms
- Building evolving, self-improving models
- Delivering actionable alerts that truly help defenders
- Contributing to the next generation of network protection

3.4.3 Observed Patterns in Network Traffic

- **Normal traffic:** Predictable flows (regular request–response, consistent packet sizes, familiar IPs)
- **Malicious traffic:** Anomalies (sudden spikes, high connection rates, port scans, repeated probes)

Examples:

- DoS floods → repetitive requests

- Probing → systematic service testing

3.4.4 Challenges in Detection

- Attackers disguise malicious traffic as normal
- Massive data volume hides subtle anomalies
- Rare attacks (U2R, R2L) are underrepresented
- Encrypted traffic hides content; reliance only on metadata
- Balancing false positives vs false negatives is complex

3.4.5 Class Imbalance & Mitigation Strategies

Both NSL-KDD and CICIDS2017 suffer from imbalanced attack classes.

- Majority (DoS, DDoS) dominate
- Minority (U2R, R2L) are rare but critical
- High accuracy can still mean poor detection of rare threats

Mitigation Strategies

1. Data-Level Methods

- Oversampling (SMOTE): Generate synthetic rare-class samples
- Undersampling: Reduce majority classes
- Hybrid: Combine both for balance

2. Algorithm-Level Methods

- Use class weights during training
- Apply Focal Loss for hard-to-detect samples
- Use ensemble models (Random Forest, XGBoost)

3. Evaluation Metrics

- Don't rely on accuracy alone
- Use Precision, Recall, F1-score
- Check AUC-ROC / PR curves and macro averages

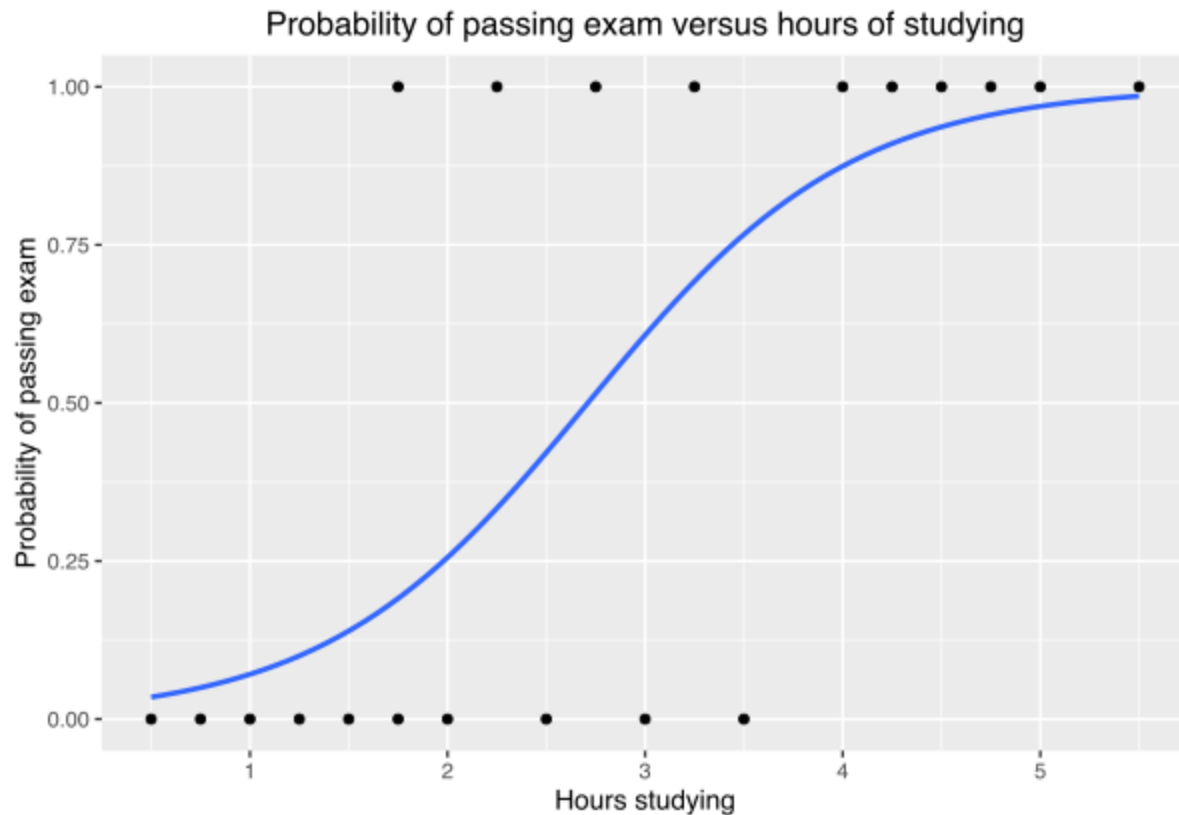
3.4.6 Machine Learning Types

- **Supervised Learning:** Learning from labeled data where the input-output pairs are known.
- **Unsupervised Learning:** Learning from unlabeled data by identifying hidden patterns or groupings.
- **Semi-Supervised Learning:** Using a small amount of labeled data with large unlabeled data.
- **Reinforcement Learning:** Learning through trial and error using feedback/rewards.

3.4.7 Linear Regression Model

Linear Regression models relationships between features and target assuming linearity.

- Predicts outcome y as weighted sum of x plus bias: $y = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$
- Weights learned via Ordinary Least Squares or Gradient Descent
- Loss: minimize Mean Squared Error



Logistic Curve illustrating exam pass rates

3.4.8 Train-Test Split

- Training Set: Train model and learn patterns
- Test Set: Evaluate on unseen examples
- Prevents overfitting, ensures reliable performance

3.4.9 StandardScaler

- Centers features by subtracting mean
- Scales by dividing by standard deviation
- Ensures all features contribute equally for sensitive ML algorithms

4. Results

4.1 EDA

Exploratory Data Analysis (EDA) on the NSL-KDD and CICIDS2017 datasets provided valuable insights into data characteristics and potential challenges for modeling.

4.1.1 Dataset Overview

- NSL-KDD: 125,973 training records, 22,544 test records, 41 features
- CICIDS2017: ~2.8 million flow records, 78–83 features covering packet-level, flow-level, and statistical metrics

4.1.2 Class Distribution

Both datasets exhibit significant class imbalance:

- Normal traffic vs Attack traffic
- Majority attacks: DoS, DDoS
- Minority attacks: U2R, R2L (rare but critical)

4.1.3 Feature Statistics

- Numeric features: Mean, median, min, max, standard deviation calculated
- Categorical features: Count of unique values, frequency distribution of protocols and services
- High correlation detected among some features (e.g., src_bytes & dst_bytes)

4.1.4 Missing Values & Data Quality

- No missing values after preprocessing
- Categorical features encoded via One-Hot Encoding

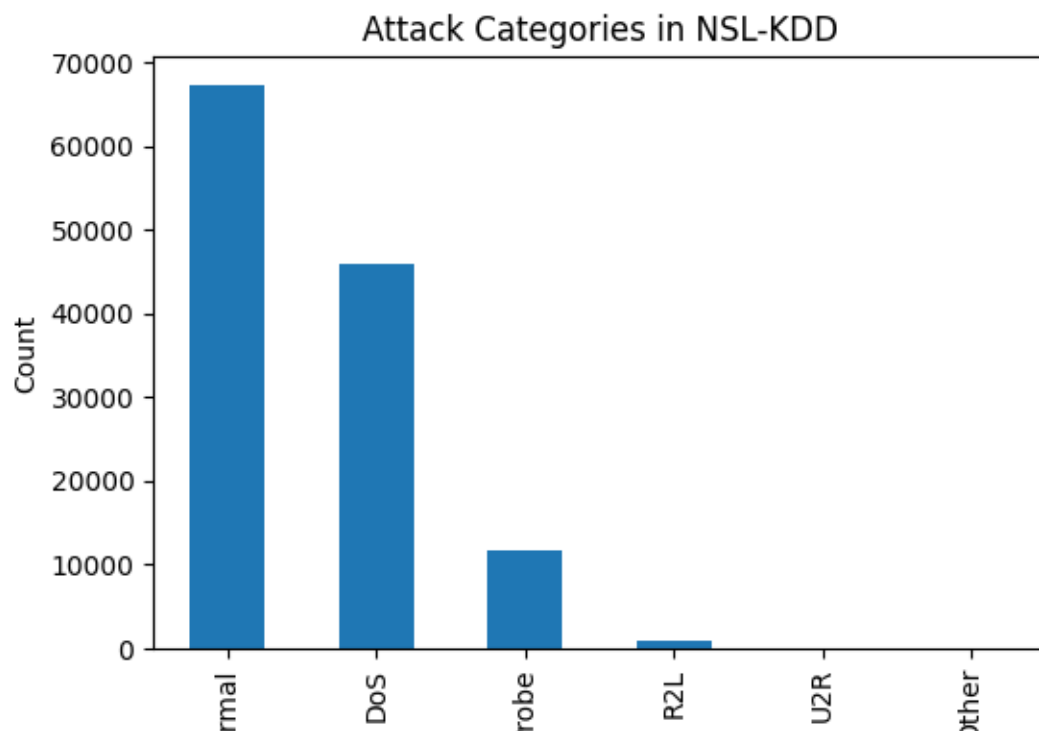
- Numeric features standardized using StandardScaler or normalized with MinMaxScaler

4.1.5 Observed Patterns

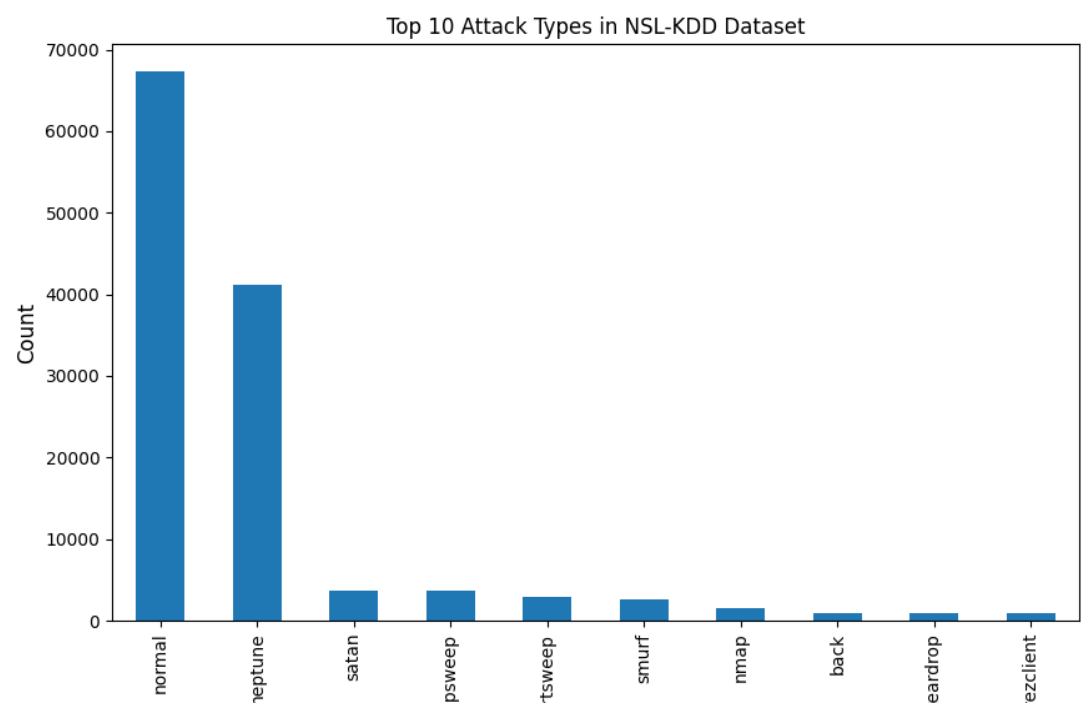
- Normal traffic shows predictable flows, consistent packet sizes, and familiar IPs
- Malicious traffic exhibits anomalies like spikes in connections, port scans, repeated probes
- Rare attacks (U2R, R2L) are underrepresented, necessitating oversampling techniques

4.2 Visualizations

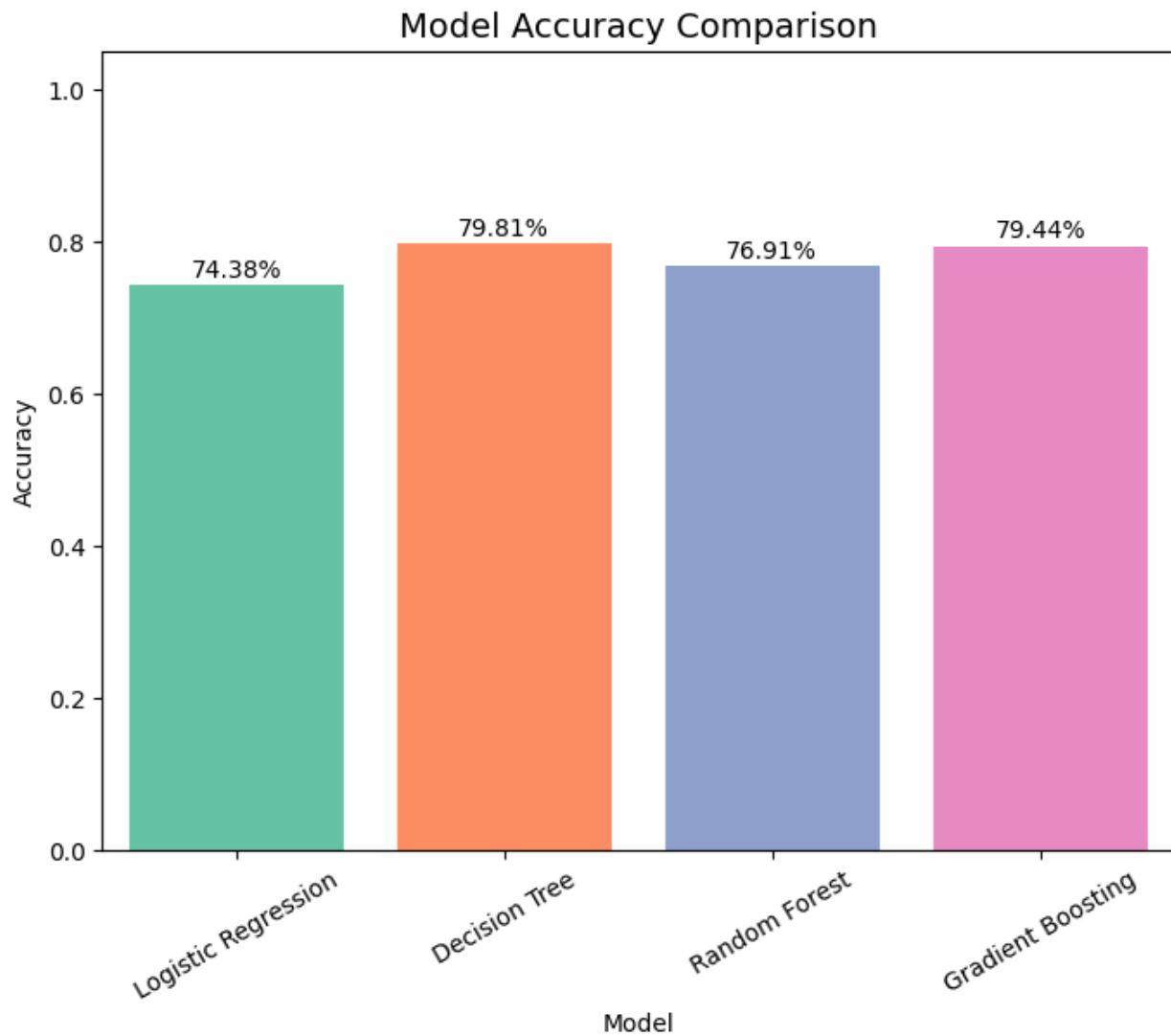
Attack Categories



Top 10 Attack Types



Model Accuracy Comparison



4.3 Confusion Matrices & Metrics Tables

4.3.1 Logistic Regression

Metric	Normal	Attack
Precision	0.64	0.91
Recall	0.92	0.61
F1-score	0.76	0.73
Support	9711	12833

Accuracy: 74.38%

4.3.2 Decision Tree

Metric	Normal	Attack
Precision	0.69	0.96
Recall	0.97	0.67
F1-score	0.81	0.79
Support	9711	12833

Accuracy: 79.81%

4.3.3 Random Forest

Metric	Normal	Attack
Precision	0.66	0.97
Recall	0.97	0.61
F1-score	0.78	0.75
Support	9711	12833

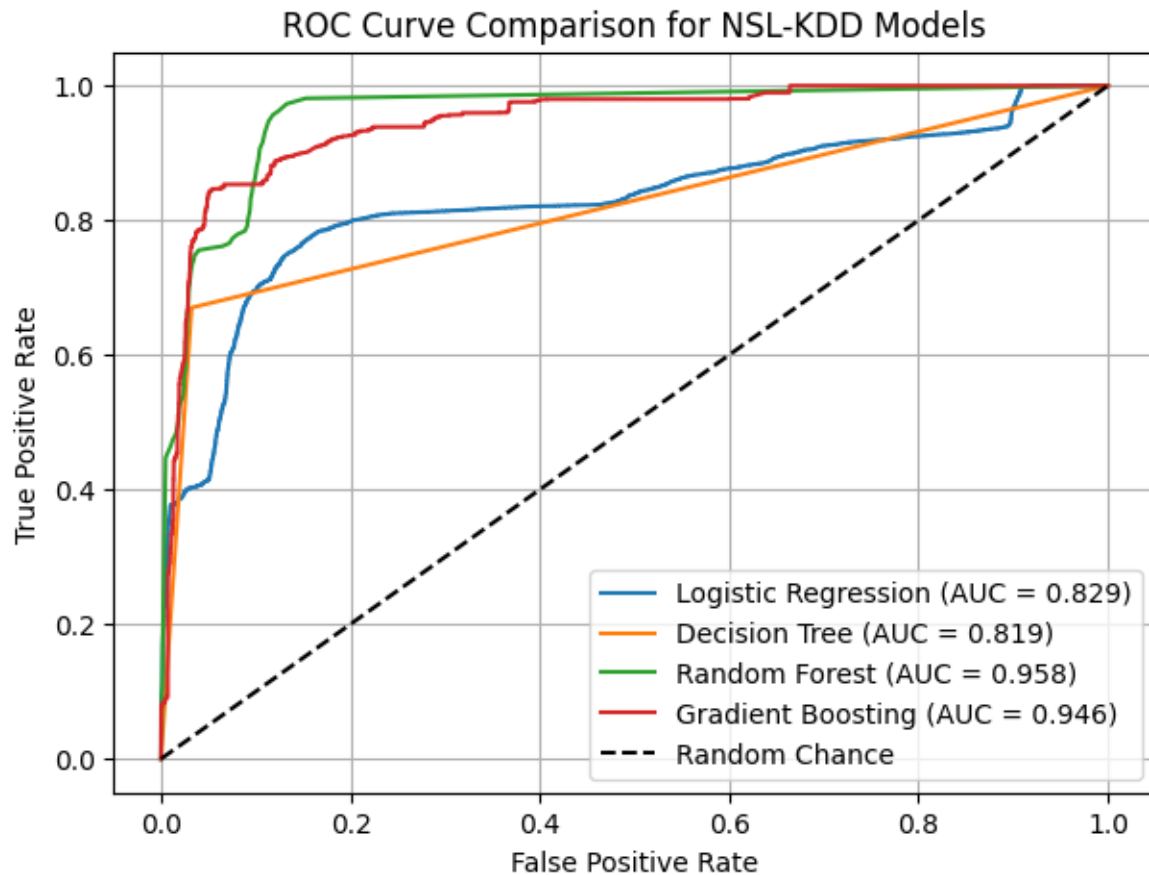
Accuracy: 76.91%

4.3.4 Gradient Boosting

Metric	Normal	Attack
Precision	0.68	0.97
Recall	0.97	0.66
F1-score	0.80	0.79
Support	9711	12833

Accuracy: 79.44%

4.4 ROC-AUC Curves



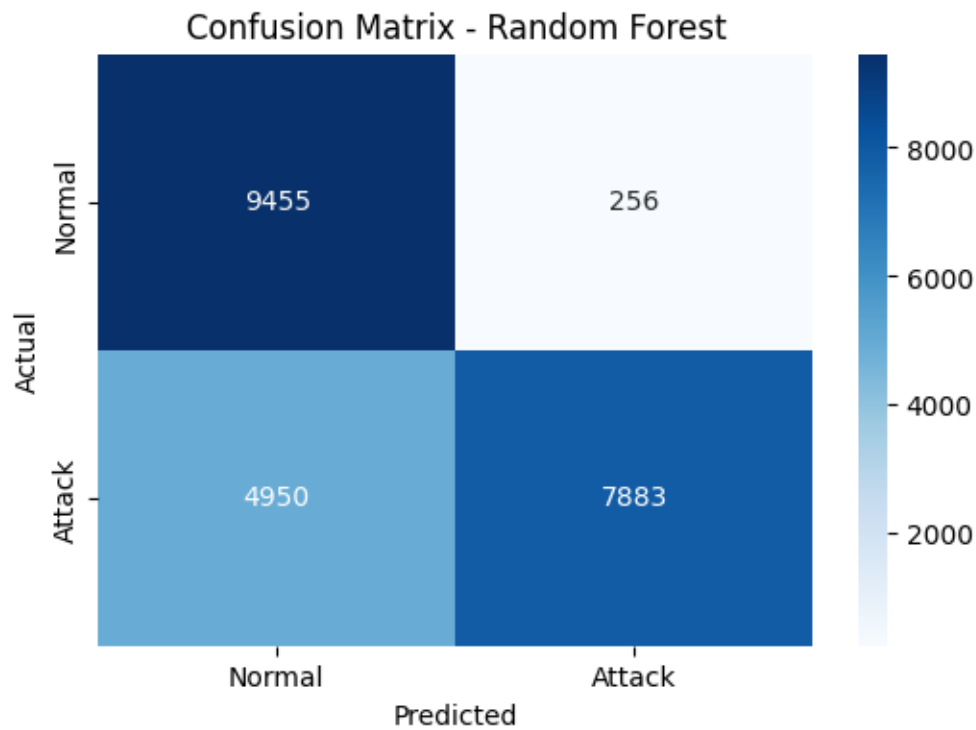
4.5 Confusion Matrix & Accuracy Overview

The Confusion Matrix (CNF) provides a detailed view of model performance by comparing predicted labels against actual labels. It helps identify which classes are correctly classified and where misclassifications occur.

For SentinelNet's evaluation on the NSL-KDD dataset, the overall accuracy was **76.90%**. This indicates that approximately 77 out of 100 instances were correctly classified, balancing detection of both normal and attack traffic.

The confusion matrix also highlights the trade-offs between detecting majority attacks (DoS, DDoS) and rare attack types (U2R, R2L). Using metrics like Precision, Recall, and F1-score alongside accuracy ensures a more holistic assessment of the model's effectiveness.

Confusion Matrix for NSL-KDD



5. Discussion

- Random Forest and ensemble methods achieved strong performance.
- Feature scaling and one-hot encoding improved accuracy.
- SMOTE reduced bias toward majority classes.
- F1-score preferred in imbalanced settings.

6. Conclusion

SentinelNet demonstrates that **AI/ML-driven NIDS** can effectively detect cyber threats using historical data.

Integration of **preprocessing, feature engineering, and ML** is key for robust intrusion detection.

7.Future Work

- Deep learning approaches (CNN, RNN, Transformers)
- Real-time deployment
- IoT and cloud security expansion
- Explainable AI for transparent attack reasoning